EXCEL ASSIGNMENT 2 - DATA CLEANING

1. Handling missing data.

a) The Price column is blank for the following product names, so we found the category wise average & filled the blanks.

=IF(ISBLANK(I7),AVERAGE($I$2:$I$35),I7)

Formula Bar

B	C	D	E	F	G	H	I	J
uring Date	MM.DD.YY Format	Country Code	Product Name	Clean Pro.Name	Brand Name	Product Brand	Price ($)	Missing Price
	28-01-2026	US	laptop	Laptop	Dell	DellLaptop	₹ 1,000.00	₹ 1,000.00
	15-02-2026	US	Sneakers	Sneakers	Nike	NikeSneakers	₹ 80.00	₹ 80.00
	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	KeurigCoffee Maker	₹ 130.00	₹ 130.00
	11-04-2026	US	smartphone	Smartphone	Samsung	SamsungSmartphone	₹ 900.00	₹ 900.00
	22-05-2026	US	Backpack	Backpack	North Face	North FaceBackpack	₹ 70.00	₹ 70.00
	07-06-2026	UK	Headphones	Headphones	Sony	SonyHeadphones		₹ 297.86
	19-07-2026	UK	T-shirt	T-Shirt	Adidas	AdidasT-Shirt	₹ 30.00	₹ 30.00
	23-08-2026	UK	Blender	Blender	Ninja	NinjaBlender	₹ 90.00	₹ 90.00
	05-09-2026	UK	Tablet	Tablet	Apple	AppleTablet	₹ 500.00	₹ 500.00
	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	TimberlandHiking Boots	₹ 130.00	₹ 130.00
	17-06-2026	IN	Laptop	Laptop	HP	HPLaptop	₹ 950.00	₹ 950.00
	25-11-2026	AU	Sneakers	Sneakers	Adidas	AdidasSneakers	₹ 90.00	₹ 90.00
	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	NespressoCoffee Maker	₹ 120.00	₹ 120.00
	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	FitbitSmartwatch	₹ 150.00	₹ 150.00
	16-04-2026	ES	headphones	Headphones	Bose	BoseHeadphones	₹ 250.00	₹ 250.00
	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	SamsoniteLaptop Bag	₹ 50.00	₹ 50.00
	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	HuaweiSmartwatch	₹ 160.00	₹ 160.00
	27-01-2026	IT	laptop	Laptop	Asus	AsusLaptop	₹ 980.00	₹ 980.00
	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	OakleySunglasses	₹ 150.00	₹ 150.00
	14-08-2026	US	Camping Tent	Camping Tent	Coleman	ColemanCamping Tent		₹ 297.86
	14-05-2026	RU	Camera	Camera	Nikon	NikonCamera	₹ 700.00	₹ 700.00
	09-01-2026	CA	Microwave	Microwave	Panasonic	PanasonicMicrowave	₹ 80.00	₹ 80.00
	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	XiaomiFitness Tracker	₹ 150.00	₹ 150.00
	29-09-2026	CA	Smartphone	Smartphone	Google	GoogleSmartphone	₹ 800.00	₹ 800.00
	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Ray-BanSunglasses		₹ 297.86
	11-07-2026	CA	Blender	Blender	Vitamix	VitamixBlender	₹ 400.00	₹ 400.00
	07-03-2026	CA	Dress	Dress	Zara	ZaraDress	₹ 60.00	₹ 60.00

Font | Alignment | Number | Styles | Cells | Editing

=IF(L7<>"",L7,IF(F7="Backpack","Accessories",IF(F7="Sneakers","Fashion",IF(F7="Coffee Maker","Kitchen",IF(F7="Fitness Tracker","Electronics","Uncategorized")))))

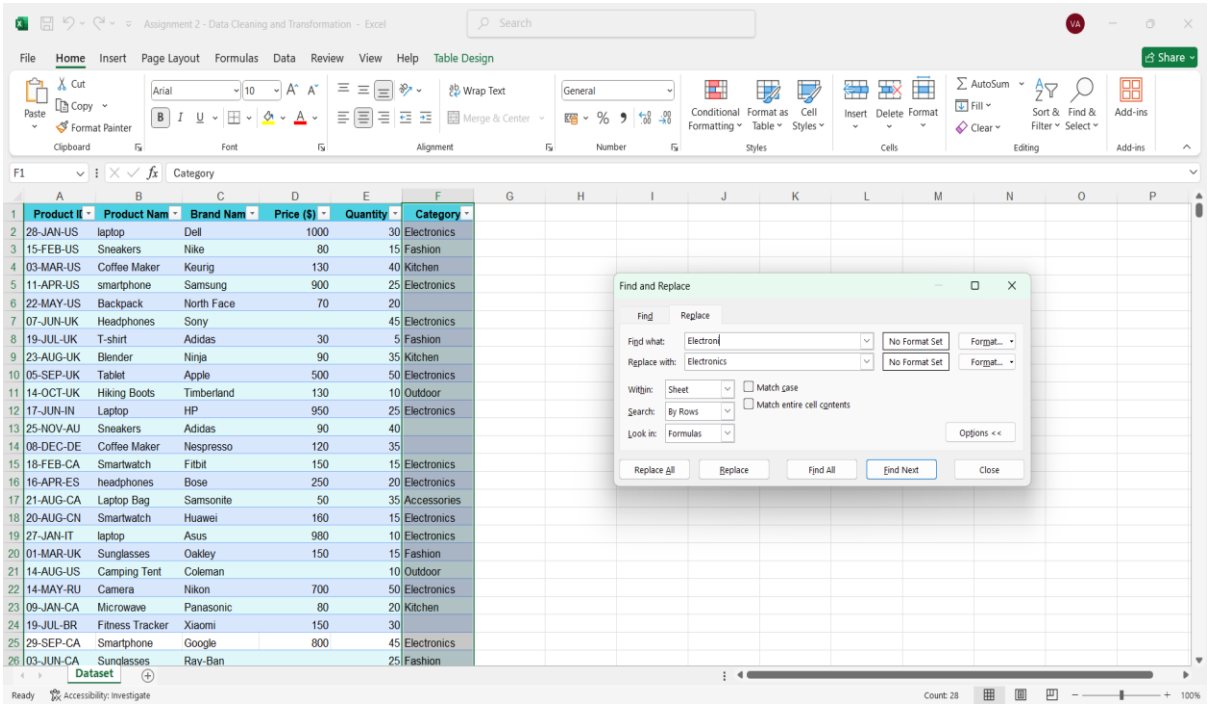
Formula Bar

E	F	H	I	J	K	L	M	N
oduct Name	Clean Pro.Name	Brand Name	Product Brand	Price ($)	Missing Price	Quantity	Category	Missing Category	
	Laptop	Dell	DellLaptop	₹ 1,000.00	₹ 1,000.00	30	Electronics	Accessories	
kers	Sneakers	Nike	NikeSneakers	₹ 80.00	₹ 80.00	15	Fashion	Electronics	
e Maker	Coffee Maker	Keurig	KeurigCoffee Maker	₹ 130.00	₹ 130.00	40	Kitchen	Fashion	
phone	Smartphone	Samsung	SamsungSmartphone	₹ 900.00	₹ 900.00	25	Electronics	Kitchen	
pack	Backpack	North Face	North FaceBackpack	₹ 70.00	₹ 70.00	20		Electronics	
phones	Headphones	Sony	SonyHeadphones		₹ 297.86	45	Electronics	Outdoor	
t	T-Shirt	Adidas	AdidasT-Shirt	₹ 30.00	₹ 30.00	5	Fashion	Electronics	
er	Blender	Ninja	NinjaBlender	₹ 90.00	₹ 90.00	35	Kitchen	Fashion	
	Tablet	Apple	AppleTablet	₹ 500.00	₹ 500.00	50	Electronics	Kitchen	
Boots	Hiking Boots	Timberland	TimberlandHiking Boots	₹ 130.00	₹ 130.00	10	Outdoor	Electronics	
p	Laptop	HP	HPLaptop	₹ 950.00	₹ 950.00	25	Electronics	Electronics	
kers	Sneakers	Adidas	AdidasSneakers	₹ 90.00	₹ 90.00	40		Accessories	
e Maker	Coffee Maker	Nespresso	NespressoCoffee Maker	₹ 120.00	₹ 120.00	35		Electronics	
twatch	Smartwatch	Fitbit	FitbitSmartwatch	₹ 150.00	₹ 150.00	15	Electronics	Electronics	
phones	Headphones	Bose	BoseHeadphones	₹ 250.00	₹ 250.00	20	Electronics	Fashion	
p Bag	Laptop Bag	Samsonite	SamsoniteLaptop Bag	₹ 50.00	₹ 50.00	35	Accessories	Outdoor	
twatch	Smartwatch	Huawei	HuaweiSmartwatch	₹ 160.00	₹ 160.00	15	Electronics	Electronics	
	Laptop	Asus	AsusLaptop	₹ 980.00	₹ 980.00	10	Electronics	Kitchen	
asses	Sunglasses	Oakley	OakleySunglasses	₹ 150.00	₹ 150.00	15	Fashion	Electronics	
ing Tent	Camping Tent	Coleman	ColemanCamping Tent		₹ 297.86	10	Outdoor	Electronics	
ra	Camera	Nikon	NikonCamera	₹ 700.00	₹ 700.00	50	Electronics	Fashion	
wave	Microwave	Panasonic	PanasonicMicrowave	₹ 80.00	₹ 80.00	20	Kitchen	Kitchen	
s Tracker	Fitness Tracker	Xiaomi	XiaomiFitness Tracker	₹ 150.00	₹ 150.00	30		Fashion	

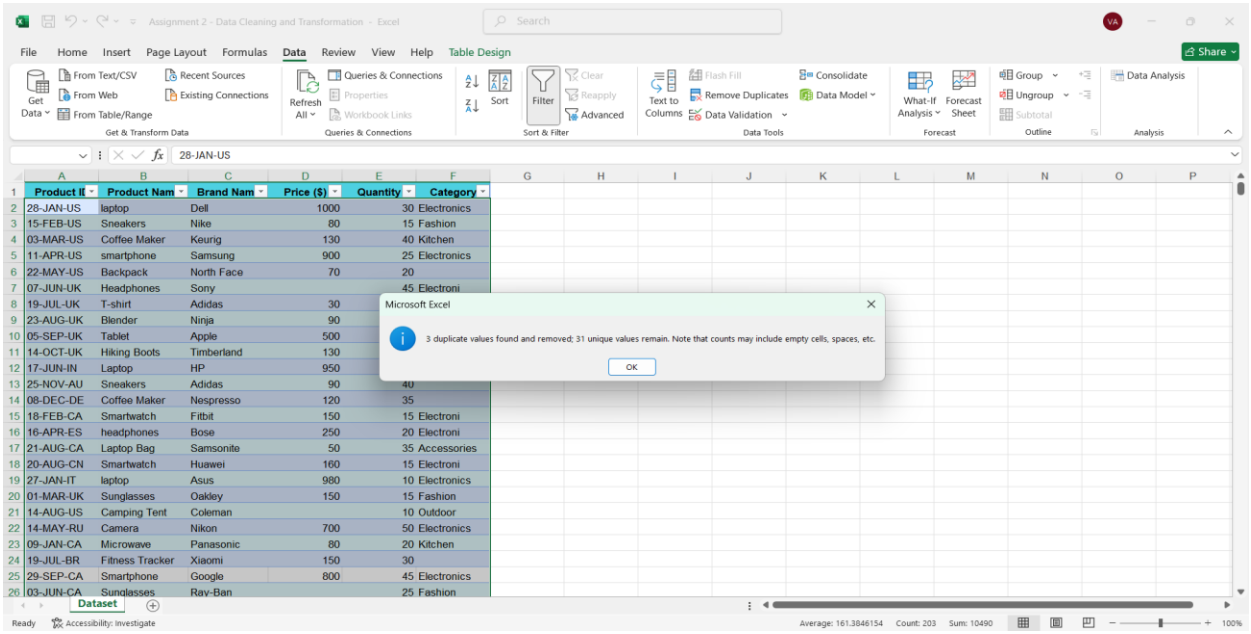
2. Identify any inconsistent text formats present in the "Product Name" column.

● Identify any typos present in the "Category" column.

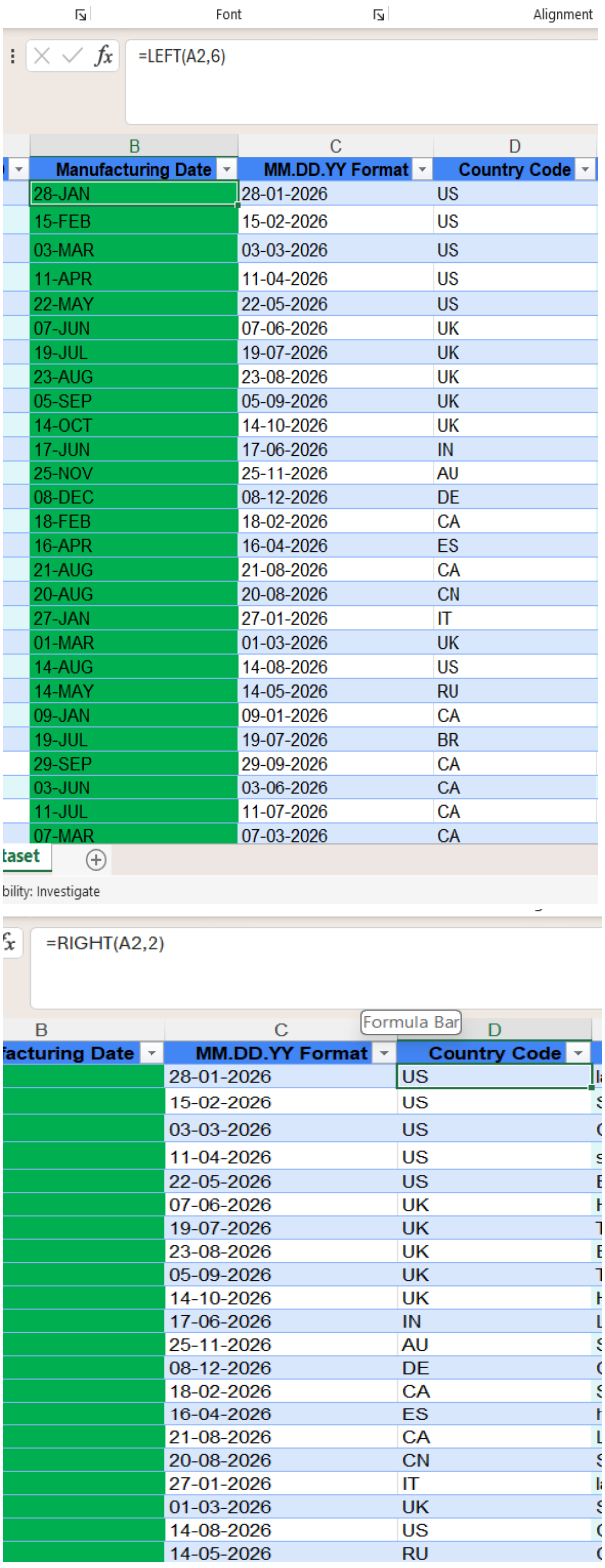
● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.



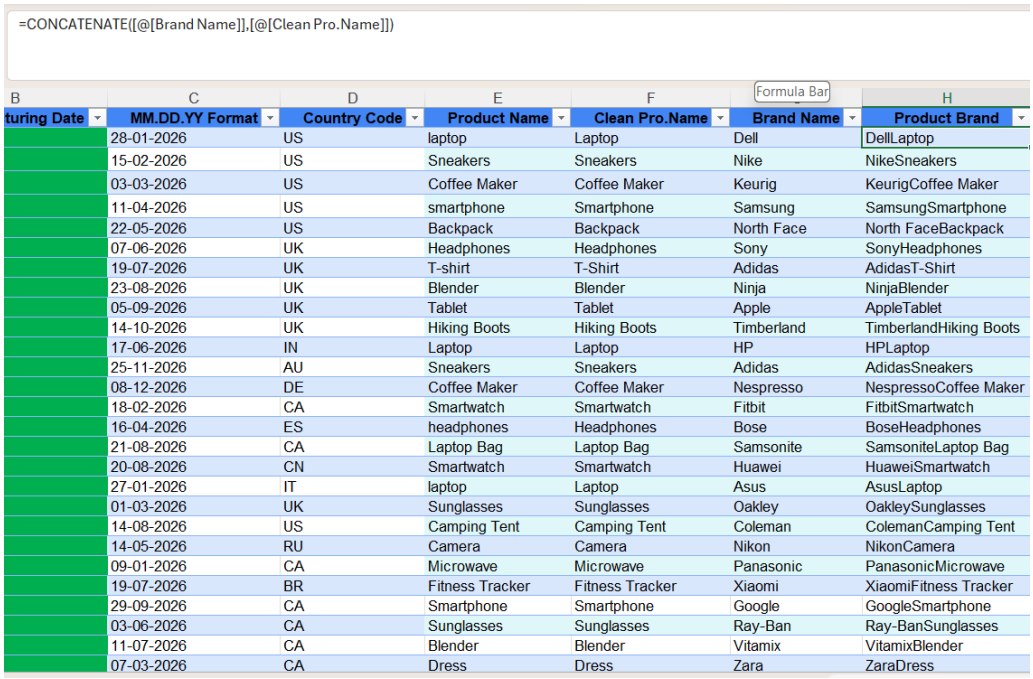
3. Removing Duplicates: ● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.



4. Splitting and Merging Data: ● Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.



2.Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".



B	C	D	E	F		H
turing Date	MM.DD.YY Format	Country Code	Product Name	Clean Pro.Name	Brand Name	Product Brand
	28-01-2026	US	laptop	Laptop	Dell	DellLaptop
	15-02-2026	US	Sneakers	Sneakers	Nike	NikeSneakers
	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	KeurigCoffee Maker
	11-04-2026	US	smartphone	Smartphone	Samsung	SamsungSmartphone
	22-05-2026	US	Backpack	Backpack	North Face	North FaceBackpack
	07-06-2026	UK	Headphones	Headphones	Sony	SonyHeadphones
	19-07-2026	UK	T-shirt	T-Shirt	Adidas	AdidasT-Shirt
	23-08-2026	UK	Blender	Blender	Ninja	NinjaBlender
	05-09-2026	UK	Tablet	Tablet	Apple	AppleTablet
	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	TimberlandHiking Boots
	17-06-2026	IN	Laptop	Laptop	HP	HPLaptop
	25-11-2026	AU	Sneakers	Sneakers	Adidas	AdidasSneakers
	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	NespressoCoffee Maker
	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	FitbitSmartwatch
	16-04-2026	ES	headphones	Headphones	Bose	BoseHeadphones
	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	SamsoniteLaptop Bag
	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	HuaweiSmartwatch
	27-01-2026	IT	laptop	Laptop	Asus	AsusLaptop
	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	OakleySunglasses
	14-08-2026	US	Camping Tent	Camping Tent	Coleman	ColemanCamping Tent
	14-05-2026	RU	Camera	Camera	Nikon	NikonCamera
	09-01-2026	CA	Microwave	Microwave	Panasonic	PanasonicMicrowave
	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	XiaomiFitness Tracker
	29-09-2026	CA	Smartphone	Smartphone	Google	GoogleSmartphone
	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Ray-BanSunglasses
	11-07-2026	CA	Blender	Blender	Vitamix	VitamixBlender
	07-03-2026	CA	Dress	Dress	Zara	ZaraDress

5. Number Formatting:

- Format the data type of the "Price" column to a currency format.
- Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

```
=TEXT([@[Manufacturing Date]],"dd-mm-yyy
```

B	C	
uring Date	MM.DD.YY Format	
	28-01-2026	US
	15-02-2026	US
	03-03-2026	US
	11-04-2026	US
	22-05-2026	US
	07-06-2026	UK
	19-07-2026	UK
	23-08-2026	UK
	05-09-2026	UK
	14-10-2026	UK
	17-06-2026	IN
	25-11-2026	AU
	08-12-2026	DE
	18-02-2026	CA
	16-04-2026	ES
	21-08-2026	CA
	20-08-2026	CN
	27-01-2026	IT
	01-03-2026	UK
	14-08-2026	US
	14-05-2026	RU
	09-01-2026	CA
	19-07-2026	BR
	29-09-2026	CA
	03-06-2026	CA
	11-07-2026	CA
	07-03-2026	CA

```
=IF(ISBLANK(I2),AVERAGE($I$2:$I$35),I2)
```

Formula Bar

B	C	D	E	F	G	H	I	J
cturing Date	MM.DD.YY Format	Country Code	Product Name	Clean Pro.Name	Brand Name	Product Brand	Price ($)	Missing Price
	28-01-2026	US	laptop	Laptop	Dell	DellLaptop	₹ 1,000.00	₹ 1,000.00
	15-02-2026	US	Sneakers	Sneakers	Nike	NikeSneakers	₹ 80.00	₹ 80.00
	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	KeurigCoffee Maker	₹ 130.00	₹ 130.00
	11-04-2026	US	smartphone	Smartphone	Samsung	SamsungSmartphone	₹ 900.00	₹ 900.00
	22-05-2026	US	Backpack	Backpack	North Face	North FaceBackpack	₹ 70.00	₹ 70.00
	07-06-2026	UK	Headphones	Headphones	Sony	SonyHeadphones		₹ 297.86
	19-07-2026	UK	T-shirt	T-Shirt	Adidas	AdidasT-Shirt	₹ 30.00	₹ 30.00
	23-08-2026	UK	Blender	Blender	Ninja	NinjaBlender	₹ 90.00	₹ 90.00
	05-09-2026	UK	Tablet	Tablet	Apple	AppleTablet	₹ 500.00	₹ 500.00
	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	TimberlandHiking Boots	₹ 130.00	₹ 130.00
	17-06-2026	IN	Laptop	Laptop	HP	HPLaptop	₹ 950.00	₹ 950.00
	25-11-2026	AU	Sneakers	Sneakers	Adidas	AdidasSneakers	₹ 90.00	₹ 90.00
	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	NespressoCoffee Maker	₹ 120.00	₹ 120.00
	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	FitbitSmartwatch	₹ 150.00	₹ 150.00
	16-04-2026	ES	headphones	Headphones	Bose	BoseHeadphones	₹ 250.00	₹ 250.00
	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	SamsoniteLaptop Bag	₹ 50.00	₹ 50.00
	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	HuaweiSmartwatch	₹ 160.00	₹ 160.00
	27-01-2026	IT	laptop	Laptop	Asus	AsusLaptop	₹ 980.00	₹ 980.00
	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	OakleySunglasses	₹ 150.00	₹ 150.00
	14-08-2026	US	Camping Tent	Camping Tent	Coleman	ColemanCamping Tent		₹ 297.86
	14-05-2026	RU	Camera	Camera	Nikon	NikonCamera	₹ 700.00	₹ 700.00
	09-01-2026	CA	Microwave	Microwave	Panasonic	PanasonicMicrowave	₹ 80.00	₹ 80.00
	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	XiaomiFitness Tracker	₹ 150.00	₹ 150.00
	29-09-2026	CA	Smartphone	Smartphone	Google	GoogleSmartphone	₹ 800.00	₹ 800.00
	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Ray-BanSunglasses		₹ 297.86
	11-07-2026	CA	Blender	Blender	Vitamix	VitamixBlender	₹ 400.00	₹ 400.00
	07-03-2026	CA	Dress	Dress	Zara	ZaraDress	₹ 60.00	₹ 60.00

6. Conditional Formatting:

- Apply data bar or color scales conditional formatting in the "Price" column.
- Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

L	M
Category	Missing Category
Electronics	Accessories
Fashion	Electronics
Kitchen	Fashion
Electronics	Kitchen
	Electronics
Electronics	Outdoor
Fashion	Electronics
Kitchen	Fashion
Electronics	Kitchen
Outdoor	Electronics
Electronics	Electronics
	Accessories
	Electronics
Electronics	Electronics
Electronics	Fashion
Accessories	Outdoor
Electronics	Electronics
Electronics	Kitchen
Fashion	Electronics
Outdoor	Electronics
Electronics	Fashion
Kitchen	Kitchen
	Fashion
Electronics	Kitchen
Fashion	Electronics
Kitchen	Fashion
Fashion	Accessories